

Overview

The Ascot Micro-Forest Connect Project seeks to establish infrastructure for tracking the growth of, and demonstrating the impact of micro-forests in the Los Angeles area and around the world.

What is a micro-forest? A micro-forest is an area of dense native plant life that mimics a naturally-occurring ecosystem. Micro-forests encourage biodiversity, manage erosion, lower air temperatures, and absorb carbon dioxide, among many other benefits!

Technologies



Firebase - Database
React - Frontend Structure
Vercel, Expo - App Deployment
MapLibre - Mapping Support

Our Team



Nolan Nguyen
Lead Developer



Jackson Muller
Contributor



Nick Kasten
Contributor

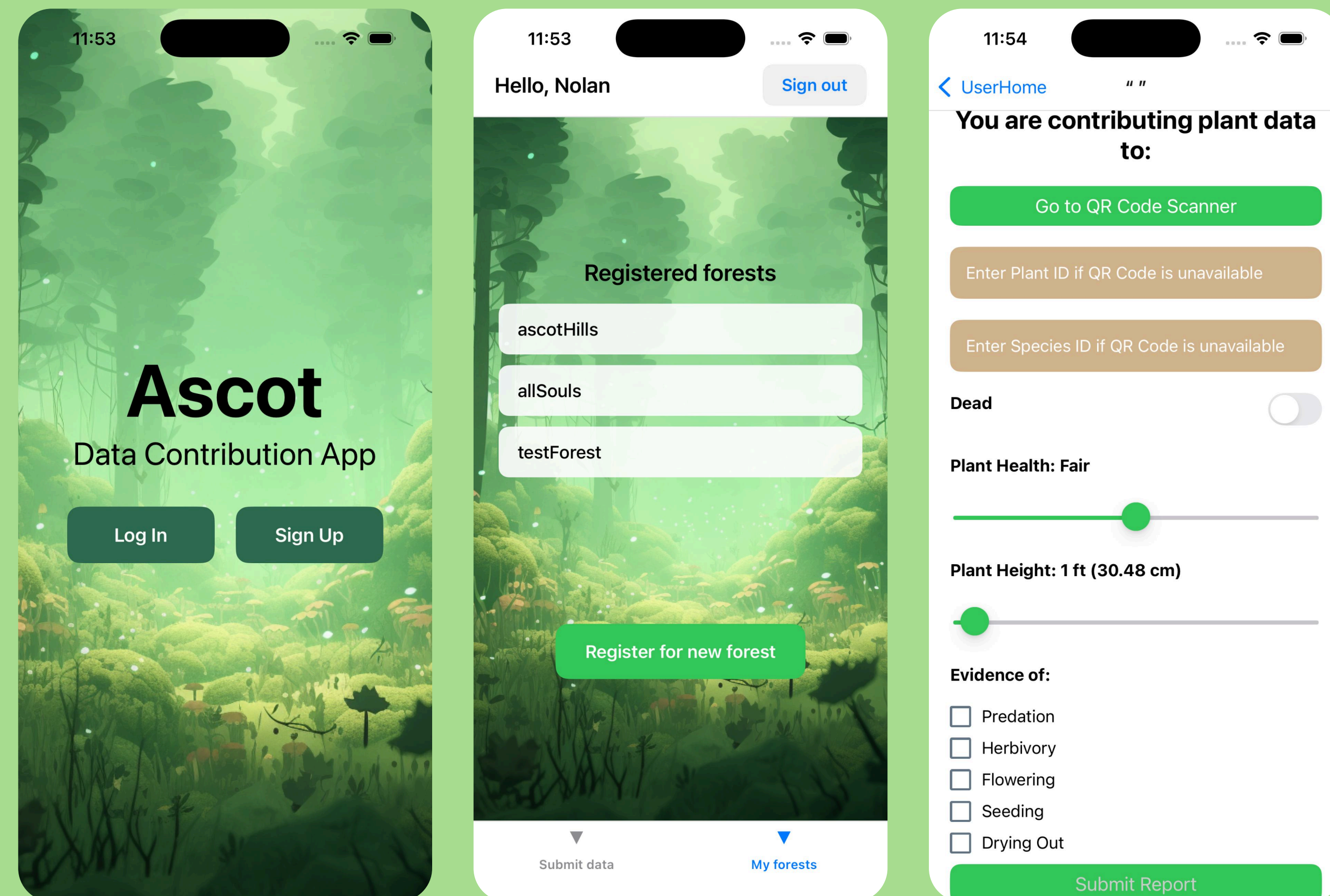
Past Devs

Christina Choi '24
Coby Sanders '24
Aaron Floreani '24
Evan Yu '24

Thank you for your contributions! :)

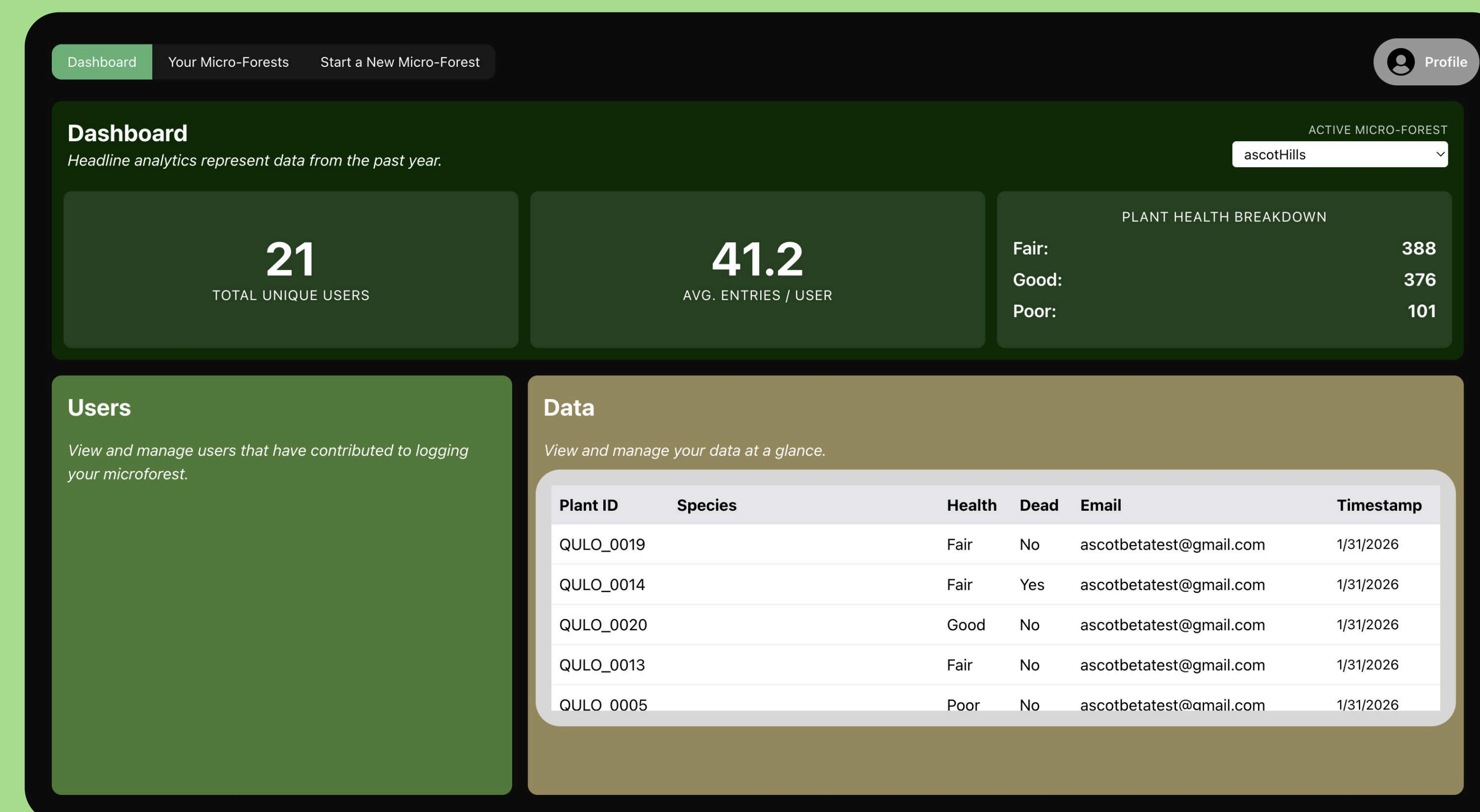
Data Contribution App

For contributors: A mobile app that allows researchers and visitors alike to record plant growth for individual micro-forests.



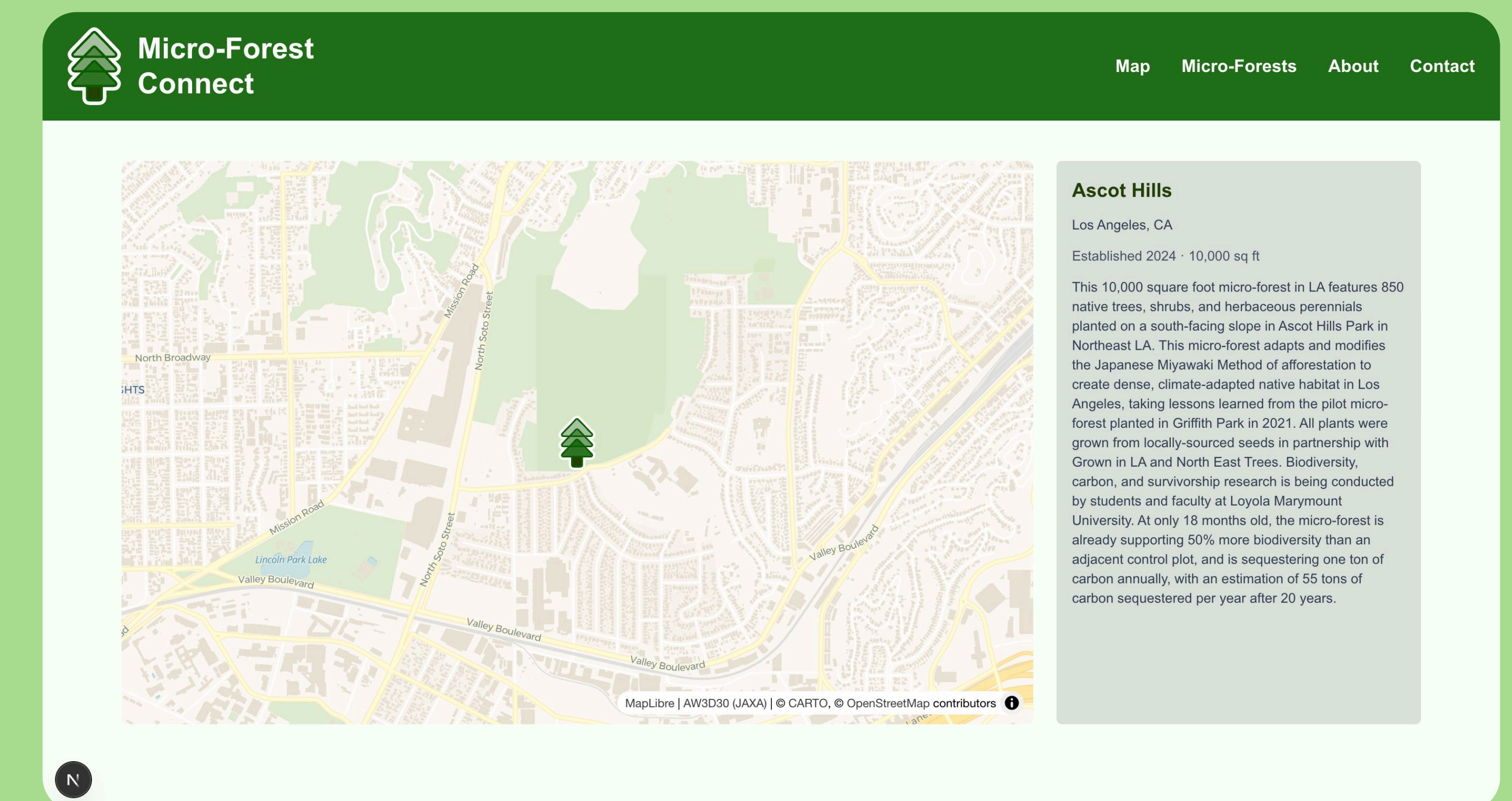
Steward Portal

For stewards, or micro-forest administrators: a portal that allows them to view statistics at a glance and manage contributors and data.



Micro-Forest Connect

For observers: an interactive web app that shows micro-forest locations and info on a dynamic map, with quick facts demonstrating impact.



Why Ascot?

I've always wanted to collaborate with a party outside of the computer science department, on a project that leverages my technical expertise with a subject matter that is tangible, physical, and new; a project centered around micro-forests checks those boxes quite nicely! I also feel compelled by the project's mission to improve education, spread awareness, and increase the impact of micro-forests in the surrounding area. Developing this software is a way for me to contribute by doing what I do best.

Usage

The project serves three different user bases: contributors, stewards and observers. **Contributors** can scan QR codes on plants in participating micro-forests to report data on their health and growth. **Stewards** have big-picture statistics at their fingertips, to inform important decisions for their micro-forests. Curious **observers** can visit an interactive, educational web page to learn about the project's impact and find ways to get involved!